Post Cardiac Arrest Management (ROSC)

ROSC Achieved

Initial Stabilization Phase

Assess Airway

Consider placement of ET Tube — P

Assess Breathing

- Start at 10 breaths/min (for adults)
- Titrate Oxygen:
  - Sp02: 92-98%
  - ETCO2: 35-45mmHg

Take Set of Vital Signs

12-Lead ECG — EMT

Assess Circulation

Blood Pressure Targets
Adults:
- SBP >90mmHg or/
- MAP >65mmHg

Pediatric:
- SBP > 5th percentile for age

Consider Crystalloid Fluid Bolus
Adult: 250ml IV/IO
Pediatric: 10ml/kg IV/IO — EMT

Continued Management And Additional Emergent Activities

Consider Vasopressor — AP

Adrenaline

	Adult	Pediatric
Push Dose	0.01mg IV/IO Every 2-5 minutes	0.005mg/kg IV/IO Every 2-5 minutes
Infusion	0.05-0.3mcg/kg/min IV	0.05-0.3mcg/kg/min IV
Dopamine		
Infusion	5-20mcg/kg/min IV/IO	5-20mcg/kg/min IV/IO

Reconsider Reversible Causes

Targeted Temperature Management
For unresponsive patients

H's and T's
- Hypovolemia
- Hypoxia
- Hydrogen ion (acidosis)
- Hypokalemia/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

Patient Care Goals

1. Optimize oxygenation, ventilation and hemodynamics in patients achieving return of spontaneous circulation (ROSC) to support organ perfusion and neurological outcome.
2. Implement measures to prevent re-arrest.
3. Ensure rapid transport to a facility capable of advanced post-cardiac arrest care appropriate to the clinical condition (e.g., STEMI).

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Adult and Pediatric patients who have achieved ROSC following cardiac arrest, including:
    - Presence of a palpable pulse.

Exclusion Criteria

- Patients with no ROSC, requiring ongoing CPR.

Key Patient Assessment Points

1. **ROSC Confirmation**
    - Verify organized rhythm.
    - Verify presence of a palpable pulse (carotid or femoral) for at least 10 seconds to confirm ROSC.
    - Monitor end-tidal CO2 (EtCO2) if available to confirm sustained circulation (EtCO2 >40 mmHg indicates ROSC).
2. **Neurological Status Post-ROSC**
    - Evaluate level of consciousness using AVPU (Alert, Verbal, Painful, Unresponsive) or Glasgow Coma Scale (GCS).
    - Assess for focal neurological deficits (e.g., pupil asymmetry, motor weakness) indicating possible brain injury.
3. **Hemodynamic Stability**
    - Monitor for signs of hypotension (SBP <90 mmHg) or poor perfusion (e.g., cool, clammy skin, delayed capillary refill >2 seconds).
    - Assess for tachycardia or bradycardia indicating potential re-arrest or underlying cardiac issues.
4. **12-Lead ECG**
    - Perform a 12-Lead ECG in all medical cardiac arrests.
    - Assess for potential cardiac causes of cardiac arrest (e.g., STEMI)
5. **Re-Arrest Risk**
    - Identify factors increasing risk of re-arrest (e.g., persistent arrhythmias, hypoxia, or hypotension).
    - Reassess for reversible causes of initial arrest (H's and T's: Hypovolemia, Hypoxia, Hydrogen ion [acidosis], Hypo-/Hyperkalemia, Hypothermia, Tension pneumothorax, Tamponade, Toxins, Thrombosis).
6. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.
        - All patients with ROSC following cardiac arrest are Red Priority.

Patient Management & Interventions

1. **Consider ALS back-up**
    - Consider the requirement for ALS back-up as per Universal Care CPG if not already activated.
    - It is appropriate to meet ALS back-up en route.
2. **Primary Survey**
    - Ensure airway patency, adequate breathing, and circulation.
    - Prepare for immediate re-initiation of CPR if re-arrest occurs.
3. **Secondary Survey**
    - Conduct a focused examination to assess neurological and hemodynamic status.
    - Obtain history from bystanders or medical records (e.g., cause of arrest, duration, and interventions performed).

- Reassess vital signs including:
  - Respiratory rate, heart rate, blood pressure, EtCO2

4. **Establish IV Access**
   - Establish IV access if not already obtained.
   - IV access should be attempted in preparation for cardiac arrest.
   - Consider placing a second IV line if vasopressors administration is anticipated.
   - Consider intraosseous (IO) access if IV access cannot be obtained.
5. **12-Lead ECG**
   - Perform a 12-lead ECG for all medical cardiac arrests.
   - 12-Lead ECG in ROSC following traumatic cardiac arrest is only required if cardiac involvement is suspected.
7. **Oxygenation and Ventilation**
   - Titrate oxygen to maintain SpO2 94-98% to avoid hyperoxia.
   - Provide ventilatory support (via BVM) if spontaneous breathing is inadequate
   - Target EtCO2 35-45 mmHg to avoid hypo- or hypercapnia.
8. **Fluid Administration**
   - Administer Normal Saline or Ringers Lactate for hypotension:
     - Adults: 250-500 mL bolus IV/IO
     - Pediatrics: 10-20mL/kg IV/IO
     - Reassess after each fluid bolus, repeat cautiously to maintain:
       - Adults: SBP ≥90 mmHg.
       - Pediatrics: SBP > 5th percentile for age
     - Avoid fluid overload in suspected cardiogenic shock (e.g., pulmonary edema).
9. **Hemodynamic Support**
   - Indicated for shock which is unresponsive to fluid administration, or if volume overload is present.
   - Titrate to maintain:
     - Adults: SBP ≥90 mmHg.
     - Pediatrics: SBP > 5th percentile for age
   - Adrenaline is generally preferred for hypotensive shock following cardiac arrest due to dopamine having a higher risk or arrhythmias at the vasopressor dose.
   - Dopamine is generally preferred for normotensive shock.
   - Dopamine
     - Adults & Pediatric: 5-20 mcg/kg/min IV/IO infusion.
   - Adrenaline
     - Indicated for shock which is unresponsive to fluid administration, or if volume overload is present.
       - Adult:
         - Push Dose: 0.01mg/kg IV/IO, every 2-5 minutes.
         - Infusion: 0.05-0.3mcg/kg/min IV/IO
       - Pediatric:
         - Push Dose: 0.005mg/kg IV/IO, every 2-5 minutes.
         - Infusion: 0.05-0.3mcg/kg/min IV/IO
10. **Arrhythmia Management**
    - Monitor for recurrent VF/VT or bradycardia; prepare for defibrillation or pacing if needed.
11. **Targeted Temperature Management (TTM)**
    - Monitor the patient's temperature to avoid hypo/hyperthermia.
    - TTM should be initiated as soon as possible
    - The target of TTM is 32-36°C
    - TTM is difficult to achieve the ambulance setting. Remove all patient clothing and ensure the vehicle air conditioning is as cold as possible.
    - Active cooling with ice packs is not routinely recommended due to the inability to adequately monitor core temperature.

12. Sedation

- Indications:
  - Patient agitation which is inhibiting patient care or patient safety.
  - Intubated patients, IV Ketamine or a Benzodiazepine may be used ensure the on-going management of an ET Tube that is already in place and ensure ventilator synchrony.
- Narcotic analgesia should not be administered to facilitate placement of an ET Tube.
- Sedation should not be administered to patients to ensure the on-going management of Supraglottic Airways (SGA's). If the SGA or OPA is causing significant agitation, it may be removed.
- Midazolam
  - Adults: 1-5mg IV, repeat as required after 10 minutes.
  - Pediatric: 0.1mg/kg IV, repeat as required after 10 minutes.
- Diazepam
  - Adults: 1-5mg IV, repeat as required after 10 minutes.
  - Pediatric: 0.1mg/kg IV, maximum 5mg, repeat as required after 10 minutes.
- Ketamine
  - Adults: 0.25-1mg/kg IV/IO, repeat as required after 10 minutes.
  - Pediatrics: 0.5-1mg/kg IV/IO, repeat as required after 10 minutes.

Patient Safety Considerations

- Hemodynamic Management
  - Avoid aggressive fluid boluses in cardiogenic shock (e.g., pulmonary edema, jugular vein distension); monitor for fluid overload.
  - Use vasopressors if hypotension persists after adequate fluid administration.
- Neurological Protection
  - Avoid hyper/hypoventilation. target EtCO2 35-45 mmHg. Use a ventilator for ventilatory support if available.
  - Monitor for seizures or neurological deterioration; administer benzodiazepines anticonvulsants as per seizures CPG if indicated.
- Transport Safety
  - A supra glottic airway should be placed for patients who remain unresponsive after cardiac arrest and are being transported.

Clinical Insights & Additional Guidance

- Avoidance of Rough Movement
  - Rapid/ rough movement of the patient should be avoided for up to 10 minutes following ROSC, as this may result in re-arrest.
- Actions Prior to Patient Movement
  - Generally a full reassessment of the patient should be undertaken prior to patient movement, including:
    - Reassessment of Airway, Breathing, Circulation and vital signs.
- Pediatric Considerations
  - Cardiac Arrest in pediatrics is typically caused by hypoxia. A strong focus must be placed on ensuring effective ventilations.
  - Intubation in pediatrics is typically more difficult than adults – there may be no need to intubate the patient if a supraglottic airway is effective.
- ROSC after Traumatic Brain Injuries
  - Maintain SBP > 90 mmHg/MAP >65 mmHg or MAP of 70-80 for isolated TBI in adults and according to age appropriate SBP range in pediatrics using (age x 2) + 70.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Description of the cardiac arrest event, timing, and circumstances of ROSC.
- Injuries / Information: Reversible causes identified (H's & T's), suspected etiology of arrest.

- Signs: ROSC time, vital signs post-ROSC (BP, HR, RR, SpO_2, $EtCO_2$), neurological status (AVPU or GCS), ECG findings.
- Treatment:
  - Interventions provided post-ROSC (oxygen, fluids, vasopressors, sedation, ventilation).
  - 12-lead ECG findings and targeted temperature management (TTM) if initiated.
- Allergies: Any relevant allergies to medications given during or after resuscitation.
- Medications: List of medications administered post-ROSC (adrenaline, dopamine, midazolam, ketamine, etc.).
- Background: Relevant history (e.g., cardiac disease, previous arrests, comorbidities).
- Other: Transport decisions, re-arrest risk, communication with receiving facility.

Key Performance Indicators

1. Perform and document a set of vital signs including $EtCO_2$ within 5 minutes of ROSC being achieved.
2. Perform a 12-lead ECG for all non-traumatic ROSC patients within 5 minutes of ROSC being achieved.

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